

Day 1: Introduction to Networking and the OSI Model

Task 1 — Wireshark Packet Capture

I used **Wireshark** to capture network traffic from my main network interface. After starting the capture, I opened a website and then stopped the capture.

In the packet list, I could see different protocols such as:

- TCP
- UDP
- DNS
- TLS
- ARP
- ICMP
- QUIC

Note: The protocols visible can be different depending on the website and network connection.

Result:

Wireshark helped me observe the actual network packets being transmitted and received by my computer.

Task 2 — Identify 3 Packets and OSI Layers

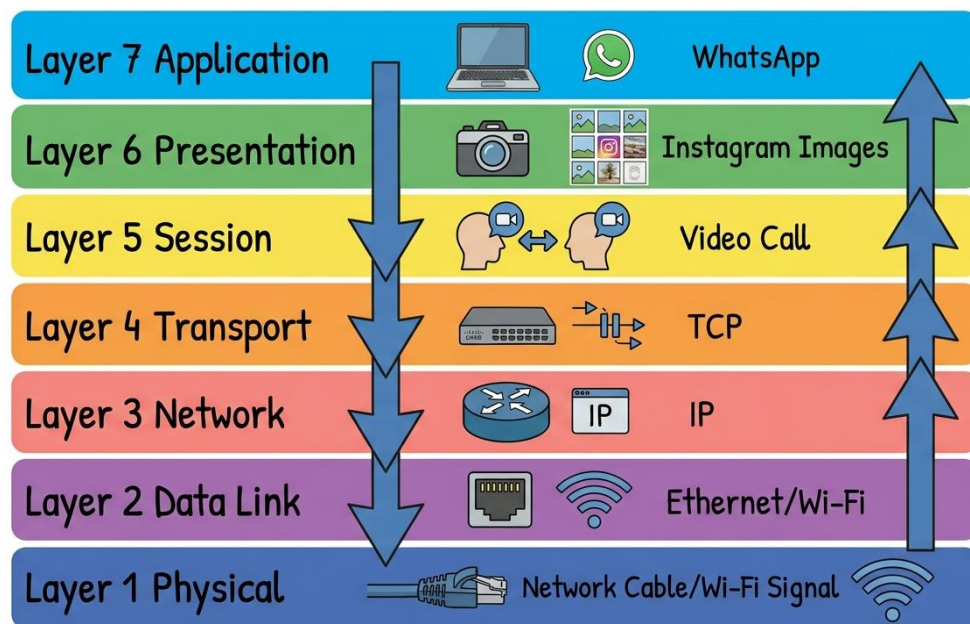
I selected three packets from the Wireshark capture and mapped them to the OSI Model

Packet	Protocol	Main OSI Layer	Reason
1	DNS	Layer 7 – Application	DNS is used to find the IP address of a domain name
2	TCP	Layer 4 – Transport	TCP provides reliable communication between devices
3	ARP	Layer 2 – Data Link	ARP is used to find the MAC address associated with an IP address on a local network

Result:

By checking the Protocol and Info columns in Wireshark, I was able to understand how different protocols are related to different OSI layers.

Task 3 — OSI Model Diagram



OSI Model Examples

Layer	Name	Example
7	Application	WhatsApp
6	Presentation	Instagram Images
5	Session	Video Call
4	Transport	TCP
3	Network	IP
2	Data Link	Ethernet / Wi-Fi
1	Physical	Network Cable / Wi-Fi Signal

Task 4 — WhatsApp Message Through OSI Model

When I send a message on WhatsApp, the data passes through different networking layers before reaching the other person's device.

1. Application Layer

I type a message in WhatsApp and press the send button. The application handles the message and prepares it for communication.

2. Transport Layer

The data is divided and managed for transmission. Transport protocols help provide communication between the sender and receiver.

3. Network Layer

The data is given IP addressing information so that it can travel through different networks towards the destination.

4. Data Link Layer

The data is prepared for communication on the local network, such as between my phone and the Wi-Fi router. MAC addresses are used at this level.

5. Physical Layer

Finally, the data is transmitted as signals through Wi-Fi or cables. The signals travel through the network until they reach the destination.

Simple Flow

**WhatsApp Message → Transport → IP/Network →
Wi-Fi/Data Link → Physical → Internet → Receiver**

Conclusion:

The OSI Model helps us understand how data moves through a network. Each layer has a different job, and together the layers allow messages and other data to travel from one device to another